

WHITEMAN AFB, MO, USA

WMO: 724467

Lat: 38.717N

Lon: 93.550W

Elev: 870

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.1	(c) 8.8	(d) -5.2	(e) 4.3	(f) 5.3	(g) -0.3	(h) 5.6	(i) 11.5	(j) 28.6	(k) 40.3	(l) 25.9	(m) 36.9	(n) 9.0	(o) 330	(p) 0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	96.6	76.5	93.1	75.9	90.5	75.6	80.1	89.3	78.6	88.1	77.5	86.8	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.8	149.0	83.9	76.3	141.6	83.3	74.9	135.0	82.5	44.2	89.5	42.8	88.1	41.5	86.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	21.8	19.4	-3.6	99.5	5.9	3.4	-7.8	102.0	-11.3	103.9	-14.6	105.8	-18.9	108.2
			-4.1	82.3	5.6	1.5	-8.1	83.4	-11.4	84.3	-14.6	85.2	-18.6	86.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	30.2	35.3	45.0	55.4	65.4	74.8	78.9	77.1	69.6	57.7	45.1	34.4
	DBStd	19.24	12.05	12.14	11.54	9.28	7.91	5.90	5.78	5.68	7.82	9.23	10.26	10.91
	HDD50	2070	620	430	230	50	3	0	0	0	0	33	209	495
	HDD65	4793	1079	831	622	309	95	5	0	1	42	261	599	949
	CDD50	4208	6	19	77	212	481	745	894	841	588	272	62	11
	CDD65	1458	0	0	4	21	108	301	430	377	180	35	2	0
	CDH74	14369	0	2	40	202	889	2881	4669	3735	1644	294	13	0
	CDH80	6027	0	0	2	30	238	1158	2205	1689	639	66	0	0
(14) Wind	WSAvg	9.2	10.5	10.5	10.8	10.9	8.9	8.7	7.4	7.0	7.6	8.9	9.8	9.6
(15) Precipitation	PrecAvg	42.8	1.7	2.0	2.8	4.6	5.5	5.6	4.4	3.4	4.2	3.4	3.0	2.1
	PrecMax	59.7	5.4	5.5	6.1	13.3	12.7	11.8	15.3	7.8	12.7	9.4	10.5	5.1
	PrecMin	26.6	0.3	0.2	1.1	0.9	1.3	1.2	0.4	0.9	0.7	0.4	0.6	0.3
	PrecStd	8.0	1.3	1.3	1.4	2.2	2.7	2.8	3.2	1.9	3.0	2.0	2.3	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	72.0	78.7	83.8	89.6	95.1	101.3	100.0	95.3	85.9	75.5	67.9
		MCWB	56.5	58.7	62.5	66.2	73.5	77.0	75.3	75.0	73.3	70.8	62.4	59.4
	2%	DB	59.4	64.8	73.4	79.5	85.7	91.8	97.3	95.3	90.7	81.5	70.7	61.8
		MCWB	51.1	54.1	59.9	63.5	71.2	76.2	77.2	75.9	73.6	65.9	59.0	55.4
	5%	DB	53.7	59.8	68.9	75.6	82.4	89.8	93.7	92.3	87.4	77.0	65.8	55.2
		MCWB	46.7	51.9	57.4	61.7	69.4	75.4	77.0	75.5	72.8	64.4	57.0	48.3
	10%	DB	47.2	54.2	63.4	71.8	79.3	87.0	90.6	89.2	83.6	72.8	61.2	50.8
		MCWB	41.6	47.4	53.0	59.9	67.4	74.2	76.6	74.9	70.6	62.8	53.6	45.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	61.8	66.4	69.4	76.0	80.4	81.8	82.0	78.3	74.3	65.0	62.2
		MCDB	63.5	67.7	74.7	79.4	84.9	89.3	91.1	89.3	85.6	82.2	72.6	66.0
	2%	WB	53.4	57.2	62.4	67.1	73.8	78.5	80.2	79.8	76.3	70.6	61.6	56.4
		MCDB	57.5	62.6	69.9	75.4	82.0	87.7	90.3	88.4	85.0	75.7	67.7	61.1
	5%	WB	47.1	52.3	58.8	64.5	71.7	77.2	78.9	78.2	74.7	67.7	58.5	50.5
		MCDB	52.9	58.5	66.6	71.9	79.2	86.3	89.0	87.3	83.3	73.9	64.1	53.8
	10%	WB	41.9	47.1	54.6	61.9	69.4	75.5	77.7	76.7	72.9	64.5	54.8	45.4
		MCDB	46.9	54.3	62.2	69.6	76.7	84.3	87.4	85.6	80.6	71.0	60.5	50.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	18.5	20.4	20.7	19.5	19.0	19.2	19.9	21.3	21.1	19.4	17.0
		MCDBR	26.7	26.9	27.5	26.3	22.2	21.5	22.3	23.8	23.8	24.7	25.2	24.0
		MCWBR	18.9	18.7	15.9	14.8	11.1	9.2	7.4	8.8	9.8	12.4	16.3	17.2
		MCDBR	24.4	24.4	23.9	21.5	19.7	19.5	18.8	20.0	20.5	20.1	21.8	21.4
	5% WB	MCWBR	18.7	19.2	15.6	14.1	11.3	9.9	8.5	9.5	9.5	12.6	16.5	18.0
(40) Clear-Sky Solar Irradiance	taub		0.297	0.310	0.337	0.388	0.425	0.427	0.444	0.447	0.387	0.346	0.307	0.292
	taud		2.463	2.427	2.410	2.281	2.238	2.265	2.234	2.234	2.359	2.461	2.514	2.499
	Ebn at Noon		280	291	293	282	272	271	265	261	272	274	275	274
	Edh at Noon		25	30	34	41	44	43	44	43	36	29	24	23
(44) All-Sky Solar Radiation	RadAvg		690	921	1210	1557	1803	2039	2036	1791	1526	1085	749	594
	RadStd		78	107	109	101	160	166	127	100	98	114	79	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	+1.33	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page